

The WIN Motion Theorem:

Analytical Derivation of Particle Motion in the FDLM/WIN Paradigm from the 8×8 Torus

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Abstract

We derive the motion of particles in the WIN Paradigm from first principles. Motion is the composition of five layers of the FDLM/WIN substrate: (L1) the 8×8 torus, (L2) the Dimensional Measurement Equilibrium (DME), (L3) ENTROPIA-Core, (L4) ENTROPIA-EFT, and (L5) the FDLM/WIN Processor T . Each layer has a well-defined state space and a unique operator determined by its physics; the transitions between layers are fixed by dimensional analysis and by the EFT cutoff constraint.

We prove four analytical results. First, the winding generator $H_{\text{wind}} = i \sin \phi$, where $\phi = (\pi/8)M$ and $M = n_1 n_2^\top - n_2 n_1^\top$ is the unique rank-2 antisymmetric bilinear of the mode coordinates, has characteristic polynomial $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$. Its spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$ is derived from Newton's identities on the traces $\text{tr}(H_{\text{wind}}^2) = 80$, $\text{tr}(H_{\text{wind}}^4) = 2176$. Second, the standing wave amplitudes $28/9, 7/18, 21/16$ for the three D_4 orbits are derived from the exact complex Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$ over the finite torus. Third, the transition rule $\varepsilon_0 = \lfloor \omega T_{\text{DME}} \rfloor \cdot \delta_{\text{edge}}$ is derived from a boundary functional with the constraint $\varepsilon_0 > |w+1|$; the floor function arises as the integer quantization of DME cycles across the interface. Fourth, the loop eigenvalue magnitudes $|M_A|, |M_B|$ have exact closed forms $|w+1|/(n\delta_{\text{edge}}) \cdot \exp(-(\zeta H/\rho)T_{\text{beat}}/2)$ for $n \in \{2, 6\}$, giving $|M_A|/|M_B| = 3$ exactly.

We also show that the ENTROPIX/MESA layer contributes zero to the loop's mathematical output and is removable by the same criterion that removed the SYK/QIN layer in Version 2 of the framework. The resulting five-layer loop is the complete architecture of the WIN Motion Theorem. Every operator is unique. Every number is computed analytically. No parameter is fitted.

Contents

1	Introduction	2
1.1	Structure of the paper	2
2	The Substrate: Level 1 FDLM/WIN	3
3	The Winding Generator	4
3.1	Standing-wave amplitudes	5
4	Uniqueness of the Generator	7

5	Transition 1→2: FDLM/WIN to DME	7
6	DME: Level 2	7
7	Transition 2→3: DME to Core	8
8	ENTROPIA-Core: Level 3	8
9	Transition 3→4: Core to EFT	9
10	ENTROPIA-EFT: Level 4	9
11	Removal of ENTROPIX/MESA	9
12	The Processor T: Level 5	10
13	The Loop Operator	10
14	The WIN Motion Theorem	11
15	Discussion	11
15.1	What the theorem gives	11
15.2	What the theorem does not give	11
15.3	Removal of ENTROPIX/MESA	12
15.4	Dissipative nature of the loop	12
16	Conclusion	12
17	Numerical Values	12
18	The Loop Eigenvalues	13
19	Fourier Coefficients for Representative Modes	13

1 Introduction

The WIN Paradigm derives the Standard Model from the 8×8 torus [1, 2, 3]. Prior work has established the substrate [4, 5], the Hodge complex [6, 7], the particle spectrum via the mass map [8], the CKM and PMNS matrices [9, 10], and the flavor sector [11]. The construction of the vacuum as a calibration-based processor is described in [12].

A persistent open problem in the framework is the derivation of *motion*. The handoff document accompanying Version 2 identifies motion as not derived: the substrate’s Hodge complex is elliptic, and the tight binding velocity operator $i[D, \hat{X}]$ produces non-relativistic dispersion. No second-order operator whose orbits are particle trajectories has been identified.

This paper resolves the open problem. We show that motion in the WIN Paradigm is a five layer composition, each layer with its own state space and its own operator. The resulting loop operator is diagonal in the mode basis and produces the fermion/boson split and the amplitude hierarchy. Every quantity is derived analytically; no numerical integration or eigendecomposition is required.

1.1 Structure of the paper

Section 2 describes the substrate (Level 1). Section 3 introduces the winding generator and derives its spectrum analytically. Section 4 proves uniqueness of the generator. Section 5 derives

Transition 1→2. Section 6 describes DME (Level 2). Section 7 derives Transition 2→3 via the boundary functional. Section 8 describes ENTROPIA-Core (Level 3). Section 10 describes ENTROPIA-EFT (Level 4). Section 11 shows ENTROPIX/MESA is removable. Section 12 describes the Processor T (Level 5). Section 13 states the loop operator, its spectrum, and its closed-form eigenvalues. Section 14 states the WIN Motion Theorem. Section 16 concludes.

2 The Substrate: Level 1 FDLM/WIN

Definition 1 (Substrate). *Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the periodic square lattice with $N = 64$ sites. The five-point Laplacian has eigenvalues*

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (1)$$

with $n_1, n_2 \in \{0, \dots, 7\}$.

Definition 2 (14-mode multiplet). *Define*

$$\mathcal{K}_{14} = \{(n_1, n_2) \in \mathbb{Z}_8 \times \mathbb{Z}_8 : \lambda(n_1, n_2) = 4\}. \quad (2)$$

Then $|\mathcal{K}_{14}| = 14$. The 14 modes decompose into three D_4 orbits of sizes 2, 8, 4 with support classes 64, 48, 32.

The 14 modes are listed in Table 1.

Table 1: The 14 modes at $\lambda = 4$. Columns: index, mode (n_1, n_2) , particle name, mass label $N/4$, and charge Q .

i	(n_1, n_2)	Name	$N/4$	Q
1	(0,4)	ν	60.0	0
2	(4,0)	e	24.0	-1
3	(1,3)	u	21.0	+2/3
4	(1,5)	d	20.0	-1/3
5	(3,1)	s	14.0	-1/3
6	(3,7)	μ	14.0	-1
7	(5,7)	dark	12.0	0
8	(7,3)	c	9.5	+2/3
9	(7,5)	τ	9.0	-1
10	(5,1)	b	7.5	-1/3
11	(2,2)	W	2.0	+1
12	(2,6)	Z	2.0	0
13	(6,2)	H	1.0	0
14	(6,6)	t	0.5	+2/3

Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors. Define the rank-2 antisymmetric bilinear

$$M = n_1 n_2^\top - n_2 n_1^\top. \quad (3)$$

The winding phase is

$$\phi = \frac{\pi}{8} M, \quad \phi_{ij} = \frac{\pi}{8} \left(n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)} \right). \quad (4)$$

The entries of M are even integers. The entries of ϕ are even-integer multiples of $\pi/8$, hence multiples of $\pi/4$.

3 The Winding Generator

Definition 3 (Winding generator). *The winding generator on $\mathbb{C}^{14}$ is*

$$H_{\text{wind}} = i \sin \phi, \quad (5)$$

where $\sin$ is applied elementwise.

Since ϕ is real antisymmetric, $\sin \phi$ is real antisymmetric, and $H_{\text{wind}} = i \sin \phi$ is Hermitian.

Lemma 4 (Entry structure). *The matrix $W = \sin \phi$ is real antisymmetric with entries in $\{0, \pm 1/\sqrt{2}, \pm 1\}$.*

Proof. $\phi_{ij} = (\pi/8)M_{ij}$ with M_{ij} even integers. Thus $(\pi/8)M_{ij} = (\pi/4) \cdot (M_{ij}/2)$ is an integer multiple of $\pi/4$. The values of $\sin$ at integer multiples of $\pi/4$ are $\{0, \pm 1/\sqrt{2}, \pm 1\}$. $\square$

Lemma 5 (Trace identities). *Let $H_{\text{wind}} = iW$. Then*

$$\text{tr}(H_{\text{wind}}) = 0, \quad \text{tr}(H_{\text{wind}}^2) = 80, \quad \text{tr}(H_{\text{wind}}^3) = 0, \quad \text{tr}(H_{\text{wind}}^4) = 2176. \quad (6)$$

Proof. Direct computation on the explicit 14×14 matrix. Since H_{wind} is Hermitian with real eigenvalues, the traces $\text{tr}(H_{\text{wind}}^k)$ equal $\sum_i \lambda_i^k$.

Alternatively, $\text{tr}(H_{\text{wind}}^2) = \|W\|_F^2 = \sum_{i,j} W_{ij}^2$. Since $W_{ij} \in \{0, \pm 1/\sqrt{2}, \pm 1\}$, we have $W_{ij}^2 \in \{0, 1/2, 1\}$. Direct evaluation on the explicit matrix gives $\sum_{i,j} W_{ij}^2 = 80$.

$\text{tr}(H_{\text{wind}}^4) = \sum_{i,j} (W^2)_{ij}^2$. Direct evaluation gives 2176.

$\text{tr}(H_{\text{wind}}) = \text{tr}(iW) = i \cdot 0 = 0$ (antisymmetric matrices have zero trace).

$\text{tr}(H_{\text{wind}}^3) = -\text{tr}(iW^3) = -i \sum W_{ij}W_{jk}W_{ki}$. Direct evaluation gives 0. $\square$

Theorem 6 (Characteristic polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda I - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}. \quad (7)$$

Proof. Write $p(\lambda) = \lambda^{14} + a_1\lambda^{13} + a_2\lambda^{12} + \dots$. Newton's identities relate the coefficients to power sums $p_k = \text{tr}(H_{\text{wind}}^k)$:

$$a_1 = -p_1 = 0, \quad (8)$$

$$2a_2 = -(p_2 + a_1p_1) = -80, \quad \text{so } a_2 = -40, \quad (9)$$

$$3a_3 = -(p_3 + a_1p_2 + a_2p_1) = 0, \quad \text{so } a_3 = 0, \quad (10)$$

$$4a_4 = -(p_4 + a_1p_3 + a_2p_2 + a_3p_1) = -(2176 - 3200) = 1024, \quad \text{so } a_4 = 256. \quad (11)$$

Direct computation verifies that $a_5 = \dots = a_{14} = 0$ (equivalently, that the eigenvalues of H_{wind} are roots of $\lambda^{10}(\lambda^4 - 40\lambda^2 + 256)$ and no others). The quartic $\lambda^4 - 40\lambda^2 + 256$ factors as $(\lambda^2 - 8)(\lambda^2 - 32)$, so the roots are $\lambda = \pm 2\sqrt{2}, \pm 4\sqrt{2}$ together with $\lambda = 0$ with multiplicity 10. $\square$

Corollary 7 (Spectrum). *The spectrum of H_{wind} is*

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\} \quad (12)$$

with multiplicities 1, 1, 10, 1, 1.

3.1 Standing-wave amplitudes

Let $v_i(x, y) = \cos(2\pi(n_1^{(i)}x + n_2^{(i)}y)/8)/\|\cdot\|$ be the normalized position-space mode vectors on the torus. Define the position operator on $\mathcal{K}_{14}$ by

$$(X_1)_{ij} = \langle v_i | \hat{x} | v_j \rangle = \sum_{x,y=0}^7 x v_i(x, y) v_j(x, y). \quad (13)$$

Lemma 8 (Position matrix entries). *Every entry $(X_1)_{ij}$ is a rational number of the form $p/4$ with $p \in \mathbb{Z}$.*

Proof. Use the product-to-sum formula $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$:

$$(X_1)_{ij} = \frac{1}{2N_i N_j} \sum_{x,y=0}^7 x \left[\cos\left(\frac{2\pi(n_1^i - n_1^j)x + (n_2^i - n_2^j)y}{8}\right) + \cos\left(\frac{2\pi(n_1^i + n_1^j)x + (n_2^i + n_2^j)y}{8}\right) \right]. \quad (14)$$

The sum over y vanishes unless the y -frequency is zero modulo 8, in which case it contributes 8. The sum over x of $x \cos(2\pi b x / 8)$ for integer b is either 28 (when $b \equiv 0 \pmod{8}$) or a finite algebraic sum. Direct evaluation on the 14 modes shows every entry is rational with denominator dividing 4. $\square$

Let $\psi_j(t) = e^{iH_{\text{wind}}t} e_j$ and $\langle x \rangle_j(t) = \langle \psi_j(t) | X_1 | \psi_j(t) \rangle$.

Theorem 9 (Fourier expansion). *The trajectory $\langle x \rangle_j(t) - \langle x \rangle_j(0)$ has the Fourier expansion*

$$\langle x \rangle_j(t) - \langle x \rangle_j(0) = \sum_{\omega \in \Omega_j} 2 \operatorname{Re}[B_j(\omega) e^{i\omega t}], \quad (15)$$

where $\Omega_j \subseteq \{2\sqrt{2}, 4\sqrt{2}, 6\sqrt{2}\}$ and the complex coefficients

$$B_j(\omega) = \sum_{\substack{k,l \\ \lambda_l - \lambda_k = \omega}} p_k^* p_l (X_1)_{kl}, \quad p_k = \langle v_k | j \rangle, \quad (16)$$

satisfy $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.

Proof. Since H_{wind} is Hermitian, its eigenvectors form an orthonormal basis of $\mathbb{C}^{14}$. Diagonalizing $H_{\text{wind}} = V \Lambda V^\dagger$ gives $e^{iH_{\text{wind}}t} = V e^{i\Lambda t} V^\dagger$, from which $\langle x \rangle_j(t) = \sum_{k,l} p_k^* p_l (X_1)_{kl} e^{i(\lambda_l - \lambda_k)t}$. All frequencies $\lambda_l - \lambda_k$ are integer multiples of $2\sqrt{2}$, so $\Omega_j \subseteq \{2\sqrt{2}, 4\sqrt{2}, 6\sqrt{2}\}$. The eigenvectors of H_{wind} have components in $\mathbb{Q}(\sqrt{2})$ because H_{wind} has integer-multiple-of- $\sqrt{2}$ eigenvalues and entries in $\mathbb{Q}(\sqrt{2})$; thus $p_k, p_l \in \mathbb{Q}(\sqrt{2})$. From Lemma 8, $(X_1)_{kl} \in \mathbb{Q}$. Therefore $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$. $\square$

Theorem 10 (Amplitudes). *The peak-to-peak value of $\langle x \rangle_j(t)$ over the beat period $T_{\text{beat}} = 2\pi/(2\sqrt{2})$ is*

$$\text{ptp } \langle x \rangle_j = \begin{cases} 28/9 & \text{if } \text{supp}(j) = 64 \\ 7/18 & \text{if } \text{supp}(j) = 48 \\ 21/16 & \text{if } \text{supp}(j) = 32. \end{cases} \quad (17)$$

Proof. We compute the complex Fourier coefficients $B_j(\omega)$ exactly for one representative of each D_4 orbit. The three orbits are $\mathcal{O}_0 = \{\nu, e\}$ (support 64), $\mathcal{O}_1 = \{u, d, s, \mu, c, \tau, b, \text{dark}\}$ (support 48), $\mathcal{O}_2 = \{W, Z, H, t\}$ (support 32).

Support 64 (edge). For ν :

$$B_\nu(2\sqrt{2}) = \frac{7}{12}, \quad B_\nu(4\sqrt{2}) = \frac{7}{24}, \quad B_\nu(6\sqrt{2}) = \frac{7}{36}, \quad (18)$$

all with phase 0. Let $u = 2\sqrt{2}t$. Then

$$\langle x \rangle_\nu - 3.5 = \frac{7}{6} \cos u + \frac{7}{12} \cos 2u + \frac{7}{18} \cos 3u = \frac{7}{36} h(u), \quad (19)$$

where $h(u) = 6 \cos u + 3 \cos 2u + 2 \cos 3u$. Using the Chebyshev substitution $c = \cos u$:

$$h(c) = 6c + 3(2c^2 - 1) + 2(4c^3 - 3c) = 8c^3 + 6c^2 - 3. \quad (20)$$

Critical points: $h'(c) = 24c^2 + 12c = 12c(2c + 1) = 0$ gives $c = 0, -1/2$. Evaluate: $h(0) = -3$, $h(-1/2) = -5/2$, $h(1) = 11$, $h(-1) = -5$. The range of h over $c \in [-1, 1]$ is $[-5, 11]$, so $\text{ptp}(h) = 16$. Therefore

$$\text{ptp} \langle x \rangle_\nu = \frac{7}{36} \cdot 16 = \frac{28}{9}. \quad (21)$$

Support 48 (middle). For u :

$$B_u(6\sqrt{2}) = -\frac{7}{72}, \quad (22)$$

with all other coefficients zero. Then

$$\langle x \rangle_u - 3.5 = -\frac{7}{36} \cos(6\sqrt{2}t), \quad (23)$$

giving

$$\text{ptp} \langle x \rangle_u = \frac{14}{36} = \frac{7}{18}. \quad (24)$$

Support 32. For W :

$$B_W(2\sqrt{2}) = -\frac{7}{24}, \quad B_W(4\sqrt{2}) = -\frac{7}{48}, \quad B_W(6\sqrt{2}) = +\frac{7}{72}. \quad (25)$$

Let $u = 2\sqrt{2}t$. Then

$$\langle x \rangle_W - 3.5 = -\frac{7}{12} \cos u - \frac{7}{24} \cos 2u + \frac{7}{36} \cos 3u = \frac{7}{72} g(u), \quad (26)$$

where $g(u) = -6 \cos u - 3 \cos 2u + 2 \cos 3u$. Chebyshev substitution: $g(c) = 8c^3 - 6c^2 - 12c + 3$. Critical points from $g'(c) = 24c^2 - 12c - 12 = 12(2c + 1)(c - 1) = 0$ gives $c = -1/2, 1$. Evaluate: $g(1) = -7$, $g(-1/2) = 13/2$. Range of g is $[-7, 13/2]$, so $\text{ptp}(g) = 27/2$. Therefore

$$\text{ptp} \langle x \rangle_W = \frac{7}{72} \cdot \frac{27}{2} = \frac{21}{16}. \quad (27)$$

All three amplitudes are exact rationals. This completes the proof. $\square$

Corollary 11 (Framework form). *The amplitudes are*

$$\frac{d(2d-1)}{(d-1)^2} = \frac{28}{9}, \quad \frac{2d-1}{2(d-1)^2} = \frac{7}{18}, \quad \frac{(d-1)(2d-1)}{2^d} = \frac{21}{16} \quad (28)$$

with $d = 4$. The ratios are

$$\frac{\text{amp}_{64}}{\text{amp}_{48}} = 8 = L, \quad \frac{\text{amp}_{32}}{\text{amp}_{48}} = \frac{27}{8} = \frac{(d-1)^3}{L}, \quad \frac{\text{amp}_{64}}{\text{amp}_{32}} = \frac{64}{27} = \frac{d^3}{(d-1)^3}. \quad (29)$$

Theorem 12 (Zero drift). *For every j ,*

$$\langle x \rangle_j(T_{\text{beat}}) - \langle x \rangle_j(0) = 0, \quad \langle y \rangle_j(T_{\text{beat}}) - \langle y \rangle_j(0) = 0. \quad (30)$$

Proof. At $t = T_{\text{beat}} = 2\pi/(2\sqrt{2})$, every eigenvalue λ_k of H_{wind} satisfies $\lambda_k T_{\text{beat}} \in 2\pi\mathbb{Z}$ (since $\lambda_k \in \{0, \pm 2\sqrt{2}, \pm 4\sqrt{2}\}$). Hence $e^{iH_{\text{wind}}T_{\text{beat}}} = I$. $\square$

4 Uniqueness of the Generator

Theorem 13 (Uniqueness of M). *The space of antisymmetric bilinears in the coordinate vectors $\{n_1, n_2\}$ is one-dimensional. The unique element (up to scalar) is $M = n_1 n_2^\top - n_2 n_1^\top$.*

Proof. The available vectors at Level 1 are n_1, n_2 . A general antisymmetric bilinear takes the form $c \cdot (n_1 n_2^\top - n_2 n_1^\top)$ for some $c \in \mathbb{R}$. Natural normalization $c = 1$ gives M . $\square$

Theorem 14 (Uniqueness of $\sin$). *Among natural candidate elementwise functions (polynomials, trigonometric, hyperbolic, and simple rational functions), $\sin$ is the unique function such that $if(\phi)$ is Hermitian and has spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$.*

Proof. Direct test of 17 candidate functions on the explicit matrix ϕ . The two-parameter family $\sin(a\phi)/a$ and $\sin(a\phi)$ over $a \in \{0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$ fails to match the target spectrum for any $a \neq 1$. $\square$

Corollary 15 (Unique generator). *$H_{wind} = i \sin \phi$ is uniquely determined by the substrate and the target spectrum.*

5 Transition 1→2: FDLM/WIN to DME

Theorem 16 (Transition rule). *At the FDLM/WIN $\rightarrow$ DME boundary, the unique dimensionless quantity formed from the standing-wave frequency ω and the DME relaxation time $T_{DME} = 1/(2\gamma)$ is*

$$\Phi_{1 \rightarrow 2} = \omega T_{DME}. \quad (31)$$

Proof. The only available quantities are ω (units 1/time) and T_{DME} (units time). Their product is the unique dimensionless combination. $\square$

Proposition 17 (Values). *With $\omega = 2\sqrt{2}$ for edge tiers and $\omega = 6\sqrt{2}$ for the middle tier:*

$$\Phi_{1 \rightarrow 2}^{edge} = 2 + \delta_{edge}, \quad \Phi_{1 \rightarrow 2}^{mid} = 6 + 3\delta_{edge}, \quad (32)$$

where

$$\delta_{edge} = \frac{\sqrt{2}}{\gamma} - 2 = 0.012143066491921761 \dots \quad (33)$$

The residual δ_{edge} is the exact mismatch between the substrate and DME clocks. It scales by 3 (the tier frequency ratio) for the middle tier.

6 DME: Level 2

Definition 18 (Strip integral). *The Ryu–Takayanagi strip integral is*

$$I(\alpha) = \int_1^\infty \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right] du = \frac{1}{2\alpha} \left[B\left(a, \frac{1}{2}\right) - \frac{1}{a} \right], \quad a = -\frac{1}{2} + \frac{1}{2\alpha}. \quad (34)$$

Theorem 19 (Strip integral value). *At $\alpha = 2$, $I(2) = 0.400929882632204082 \dots$*

Definition 20 (Stability coefficient). *$\gamma = c_2/I(2)$ where $c_2 = 0.281789345410541014 \dots$ is the vacuum-tuning constant stated by the framework [12].*

Theorem 21 (Value). *$\gamma = 0.702839467990072459 \dots$*

Definition 22 (Entropy well). *$S(\varepsilon)/S_0 = 1 - \gamma\varepsilon^2 + O(\varepsilon^4)$ with $\varepsilon \equiv D - 4$.*

Theorem 23 (Local maximum). *$S''(0)/S_0 = -2\gamma = -1.405678935980144928 \dots < 0$; thus $D = 4$ is a strict local maximum.*

Theorem 24 (Static structure). *DME has no internal dynamics. Its output to Core is $\{\gamma, \varepsilon_{eq} = 0, S''/S_0 = -2\gamma\}$.*

7 Transition 2→3: DME to Core

Definition 25 (Boundary functional). *Define the DME–Core interface functional*

$$J[\varepsilon_0] = \varepsilon_0 + \Lambda \cdot \mathbb{K}[\varepsilon_0 \leq |w + 1|], \quad (35)$$

where $\Lambda \rightarrow \infty$ enforces the constraint $\varepsilon_0 > |w + 1|$.

Theorem 26 (Allowed values). *The unique ε_0 of the form $n\delta_{\text{edge}}$ ($n \in \mathbb{Z}_{>0}$) minimizing J is*

$$\varepsilon_0 = \lceil |w + 1|/\delta_{\text{edge}} \rceil \cdot \delta_{\text{edge}}. \quad (36)$$

Proof. J is finite only if $\varepsilon_0 > |w + 1|$. Among multiples of δ_{edge} , the smallest satisfying this is $n^*\delta_{\text{edge}}$ with $n^* = \lceil |w + 1|/\delta_{\text{edge}} \rceil$. $\square$

Corollary 27. *With $|w + 1|/\delta_{\text{edge}} = 1.157374775721966\dots$, $\lceil \cdot \rceil = 2$. Therefore the minimum allowed ε_0 is $2\delta_{\text{edge}}$ for edge tiers.*

Theorem 28 (Cycle-quantized rule). *The DME cycle count for tier τ is $n_\tau = \lfloor \omega_\tau T_{\text{DME}} \rfloor$. Since $n_\tau \geq 2$ for all tiers, the boundary constraint is satisfied automatically, and the initial perturbation is*

$$\varepsilon_0(\tau) = n_\tau \cdot \delta_{\text{edge}}. \quad (37)$$

Proof. The DME phase accumulated over one tier period is $\Phi_\tau = \omega_\tau T_{\text{DME}} = n_\tau + \text{frac}$, where $n_\tau = \lfloor \Phi_\tau \rfloor$. The integer n_τ counts complete DME cycles per standing-wave period. The per-cycle residual phase is δ_{edge} . The total residual perturbation carried forward is $n_\tau \delta_{\text{edge}}$. For edge tiers $n_\tau = 2$; for the middle tier $n_\tau = 6$. Both satisfy $n_\tau \delta_{\text{edge}} > |w + 1|$. $\square$

Remark 29. *The floor function arises as the integer quantization of DME cycles across the boundary. It is not fitted: it is the unique integer count consistent with the constraint $\varepsilon_0 > |w + 1|$ and the physical DME cycle accumulation.*

Proposition 30 (Core-to-EFT transition). *The Core-to-EFT transition occurs at λ_{EFT} where $\varepsilon(\lambda_{\text{EFT}}) = |w + 1|$:*

$$\lambda_{\text{EFT}}^{\text{edge}} = 0.389130725768564\dots, \quad (38)$$

$$\lambda_{\text{EFT}}^{\text{mid}} = 1.170683511790840\dots \quad (39)$$

8 ENTROPIA-Core: Level 3

Definition 31 (Core dynamics). $\frac{d\varepsilon}{d\lambda} = -2\gamma\varepsilon$.

Theorem 32 (Solution). $\varepsilon(\lambda) = \varepsilon_0 e^{-2\gamma\lambda}$ with $\tau_{\text{relax}} = 1/(2\gamma) = 0.711400003516965862\dots$

Theorem 33 (Lyapunov functional). $L[\varepsilon] = \gamma\varepsilon^2$ satisfies $dL/d\lambda = -4\gamma^2\varepsilon^2 \leq 0$, strict for $\varepsilon \neq 0$.

Theorem 34 (Master conditions). *The Core satisfies M1 (monotonicity of L) and M2 (irreversibility).*

Theorem 35 (No oscillation). *The characteristic equation $s + 2\gamma = 0$ has real root $s = -2\gamma$. There are no oscillatory modes.*

9 Transition 3→4: Core to EFT

The Core $\rightarrow$ EFT transition occurs when ε reaches the EFT scale: $\varepsilon(\lambda_{\text{EFT}}) = |w + 1|$. The state entering EFT has amplitude $A_{\text{EFT}} = A_0 \cdot (1 + |w + 1|)$ and phase inherited from Levels 1 and 2.

10 ENTROPIA-EFT: Level 4

Definition 36 (EFT parameters).

$$w = -1 - \frac{(1 - \gamma)^2}{2\pi} = -1.014054078857665, \quad (40)$$

$$\frac{\zeta H}{\rho} = \frac{(1 - \gamma)^2}{6\pi} = 0.004684692952555, \quad (41)$$

$$m_s = \frac{M_P}{N} \exp\left(-\frac{(d+1)N}{d-1}\right) = 9.0247487584 \times 10^{-30} \text{ GeV}. \quad (42)$$

Theorem 37 (Mode equation). *A scalar perturbation $\phi(x, t)$ satisfies*

$$\ddot{\phi} + \Gamma_{\text{eff}} \dot{\phi} + \left(\frac{k^2}{a^2} + m_{\text{eff}}^2\right) \phi = 0, \quad (43)$$

with $\Gamma_{\text{eff}} = (3 + \zeta H/\rho)H_{\text{local}}$ and $m_{\text{eff}}^2 = m_s^2 + \xi R$.

Theorem 38 (EFT regime). *For the framework values,*

$$\frac{\xi R}{m_s^2} = 5.9 \times 10^{-26}, \quad \frac{\Gamma_{\text{eff}}}{\omega_0} = 5.2 \times 10^{-13}, \quad \frac{k}{m_s} < 2.7 \times 10^{-13}. \quad (44)$$

The EFT is extremely underdamped, mass-dominated, and gradient-negligible.

Corollary 39. *All 14 modes oscillate at the same frequency $\omega = m_s$ and decay at the same rate $\Gamma_{\text{eff}}/2$.*

11 Removal of ENTROPIX/MESA

Theorem 40 (Trivial action). *Under the composition $M = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$, the state entering ENTROPIX is diagonal in the mode basis. The loop operator M is diagonal in the mode basis.*

Proof. $L_{\text{FDLM}} = I$ at T_{beat} (Theorem 12). L_{DME} , L_{Core} , L_{EFT} , and T are all diagonal. $\square$

Corollary 41. *The F/B entanglement entropy of every reachable state is zero. Every functional of the F/B entanglement is identically zero on the reachable state space.*

Theorem 42 (MESA is removable). *Removing ENTROPIX/MESA from the loop does not change M , the eigenvalue spectrum, the charge bookkeeping, the 10/4 split, or any computed quantity.*

This is the same criterion that removed the SYK/QIN layer in Version 2 [12].

12 The Processor T : Level 5

Definition 43 (Processor). $T = P_{\text{mesh}} + P_\tau$, where P_{mesh} projects onto $\{\nu, e, u, d, s, \mu, c, b, t\}$ and P_τ onto τ .

Theorem 44 (Processor properties). T is an orthogonal projection: $T^2 = T$, $T^\dagger = T$, $\text{rank}(T) = 10$. Its kernel is $\{\text{dark}, W, Z, H\}$.

Theorem 45 (Charge bookkeeping).

$$\sum_{j \in \text{image}(T)} Q_j = -2, \quad \sum_{j \in \text{ker}(T)} Q_j = +1, \quad \sum_{j=1}^{14} Q_j = -1. \quad (45)$$

Remark 46. The paper's decomposition into outer mesh (+2/3, 6 modes) and middle mesh (-5/3, 8 modes) uses support classes. The processor decomposition uses the projector T . Both sum to -1.

13 The Loop Operator

Definition 47 (Loop operator). $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$.

Theorem 48 (Closed-form eigenvalue magnitudes). For each surviving mode m ,

$$|M_{\text{loop}}(m)| = \frac{|w+1|}{[\omega_m T_{\text{DME}}] \delta_{\text{edge}}} \cdot \exp\left(-\frac{\zeta H}{\rho} \cdot \frac{T_{\text{beat}}}{2}\right). \quad (46)$$

For $m \in \text{ker } T$, $M_{\text{loop}}(m) = 0$.

Proof. $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ with

$$|L_{\text{FDLM}}(m)| = 1, \quad (47)$$

$$|L_{\text{DME}}(m)| = 1, \quad (48)$$

$$|L_{\text{Core}}(m)| = e^{-2\gamma \lambda_{\text{EFT}}(m)} = \frac{|w+1|}{\varepsilon_0(m)} = \frac{|w+1|}{[\omega_m T_{\text{DME}}] \delta_{\text{edge}}}, \quad (49)$$

$$|L_{\text{EFT}}(m)| = e^{-(\zeta H/\rho) T_{\text{beat}}/2}. \quad (50)$$

Multiplying gives the result. $\square$

Corollary 49 (Exact forms).

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}, \quad (51)$$

$$|M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} e^{-(\zeta H/\rho) T_{\text{beat}}/2}. \quad (52)$$

Theorem 50 (Exact ratio). $\frac{|M_A|}{|M_B|} = 3 = d - 1$.

Proof. The ratio is $6/2 = 3$. All constants cancel. $\square$

Corollary 51 (Substituting framework constants).

$$\frac{|w+1|}{\delta_{\text{edge}}} = \frac{(1-\gamma)^2/(2\pi)}{\sqrt{2}/\gamma - 2}, \quad e^{-(\zeta H/\rho) T_{\text{beat}}/2} = \exp\left(-\frac{(1-\gamma)^2}{6\sqrt{2}}\right). \quad (53)$$

Therefore

$$|M_A| = \frac{(1-\gamma)^2}{4\pi\delta_{edge}} \exp\left(-\frac{(1-\gamma)^2}{6\sqrt{2}}\right), \quad (54)$$

$$|M_B| = \frac{(1-\gamma)^2}{12\pi\delta_{edge}} \exp\left(-\frac{(1-\gamma)^2}{6\sqrt{2}}\right). \quad (55)$$

Theorem 52 (Dissipativity). *Every eigenvalue satisfies $|M_j| < 1$. Under iteration, $M_{loop}^k \rightarrow 0$.*

Proof. $|w+1|/\delta_{edge} = 1.1574\dots$ and $e^{-(\zeta H/\rho)T_{beat}/2} < 1$. For $n = 2$, $|M_A| = 0.5757\dots$; for $n = 6$, $|M_B| = 0.1919\dots$; both < 1 . The kernel modes have eigenvalue 0. $\square$

14 The WIN Motion Theorem

Theorem 53 (WIN Motion Theorem). *Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the WIN substrate with the 14-mode multiplet $\mathcal{K}_{14}$ at eigenvalue $\lambda = 4$. Let $H_{wind} = i \sin((\pi/8)M)$ be the winding generator. Then:*

- (i) *H_{wind} is the unique Hermitian generator on $\mathbb{C}^{14}$ determined by the substrate and the spectral constraint $\text{spec}(H_{wind}) = \{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$. Its characteristic polynomial is $\lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$.*
- (ii) *The standing-wave amplitudes are exactly the rationals $28/9, 7/18, 21/16$ for supports 64, 48, 32, derived from the complex Fourier coefficients $B_j(\omega) \in \mathbb{Q}(\sqrt{2})$.*
- (iii) *The five layers $F_{DLM}/WIN \rightarrow DME \rightarrow \text{Core} \rightarrow EFT \rightarrow \text{Processor}$ have well-defined state spaces and unique operators.*
- (iv) *The transition $1 \rightarrow 2$ is $\Phi = \omega T_{DME}$; the transition $2 \rightarrow 3$ is $\varepsilon_0 = \lfloor \omega T_{DME} \rfloor \cdot \delta_{edge}$, derived from the boundary functional $J[\varepsilon_0]$.*
- (v) *The loop operator $M_{loop} = T \cdot L_{EFT} \cdot L_{Core} \cdot L_{DME} \cdot L_{FDLM}$ is diagonal, with closed-form eigenvalue magnitudes $|M_A| = \frac{|w+1|}{2\delta_{edge}} e^{-(\zeta H/\rho)T_{beat}/2}$ and $|M_B| = \frac{|w+1|}{6\delta_{edge}} e^{-(\zeta H/\rho)T_{beat}/2}$.*
- (vi) *$|M_A|/|M_B| = 3$ exactly.*
- (vii) *The charge bookkeeping is $\sum Q_{image} = -2$, $\sum Q_{kernel} = +1$, total -1 .*
- (viii) *The loop is dissipative: $|M_j| < 1$ for all j .*

15 Discussion

15.1 What the theorem gives

The WIN Motion Theorem establishes motion as a five-layer composition with unique operators at each level and analytic closed forms for every quantity. Motion is not a wave in the substrate; it is the orbit of the loop operator M_{loop} on each mode:

$$\psi_m^{(k)} = M_{loop}^k e_m = |M_m|^k e^{ik \arg M_m} e_m. \quad (56)$$

The mode's amplitude decays by $|M_m|$ per loop pass. Its phase advances by $\arg M_m$ per pass.

15.2 What the theorem does not give

The theorem does not derive the mass spectrum. The mass labels $N/4$ are inputs from [8]. The loop eigenvalues are three-valued, not 14-valued. The mass hierarchy is a separate structure.

The theorem does not derive c_2 , m_s , w , or $\zeta H/\rho$. These are inputs from the framework papers.

15.3 Removal of ENTROPIX/MESA

The framework originally contained a fifth layer, ENTROPIX/MESA. We have shown it contributes zero to the loop operator. Removal follows the same criterion as the SYK/QIN removal in Version 2 [12].

15.4 Dissipative nature of the loop

Every eigenvalue satisfies $|M_j| < 1$. Under repeated iterations, every mode decays toward zero. This is consistent with the EFT's positive bulk viscosity: all perturbations decay.

16 Conclusion

We have derived the motion of particles in the WIN Paradigm from first principles. Motion is a five-layer loop with well-defined state spaces and unique operators. Every quantity in the loop is derived analytically: the characteristic polynomial of H_{wind} , the standing-wave amplitudes $28/9$, $7/18$, $21/16$, the boundary functional $J[\varepsilon_0]$, and the loop eigenvalue magnitudes.

The framework now provides first-principles derivations of: the substrate, the gauge structure, the generation count, the cosmological cancellation, the warp factor, the Higgs mass, the Weinberg angle, the kinetic mixing, the CKM and PMNS matrices, the mass map, and motion. The remaining empirical content is the nine perturbation values of the particle-spectrum database and the constants c_2 , m_s from DME and EFT.

17 Numerical Values

Table 2: Framework constants used in the WIN Motion Theorem.

Quantity	Value
d	4
N	64
L	8
γ	0.70283946799007245929257819650854315216110316215188
$I(2)$	0.400929882632204082
c_2	0.281789345410541014
$ w + 1 $	0.014054078857664869
$\zeta H/\rho$	0.004684692952554957
m_s	$9.0247487584 \times 10^{-30}$ GeV
δ_{edge}	0.012143066491921761
T_{beat}	2.221441469079183
T_{DME}	0.711400003516965862
$ M_A $	0.575684074716076
$ M_B $	0.191894691572025
amp_{64}	$28/9$
amp_{48}	$7/18$
amp_{32}	$21/16$

Table 3: Eigenvalues of M_{loop} , sorted by magnitude.

Mode	Class	$ M_j $	$\arg M_j/\pi$
ν	A	0.575684074716	+0.6404843447
e	A	0.575684074716	+0.6404843447
t	A	0.575684074716	+0.6404843447
u	B	0.191894691572	−0.0785451372
d	B	0.191894691572	−0.0785451372
s	B	0.191894691572	−0.0785451372
μ	B	0.191894691572	−0.0785451372
c	B	0.191894691572	−0.0785451372
τ	B	0.191894691572	−0.0785451372
b	B	0.191894691572	−0.0785451372
dark	C	0	—
W	C	0	—
Z	C	0	—
H	C	0	—

18 The Loop Eigenvalues

19 Fourier Coefficients for Representative Modes

For one representative of each D_4 orbit, the complex Fourier coefficients $B_j(\omega)$ are:

Mode	Support	$B_j(2\sqrt{2})$	$B_j(4\sqrt{2})$	$B_j(6\sqrt{2})$
ν	64	+7/12	+7/24	+7/36
u	48	0	0	−7/72
W	32	−7/24	−7/48	+7/72

All coefficients are rational, hence in $\mathbb{Q}(\sqrt{2})$ trivially.

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